

Design and development of a digital reference verification and validation tool

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Meeting snapshot

Research direction	Verify references against trusted scholarly metadata, explain mismatches at field level, and produce a traceable corrected citation.
Working prototype	Browser interface with live Crossref DOI lookup and title search, candidate metadata, confidence display, field warnings, an evidence trace, filters, CSV export, and draft IEEE, APA and Harvard formatting.
Current limit	Confidence is an initial title-similarity heuristic and formatting is a demonstrator. A labelled corpus, multi-field scoring, calibration and CSL rendering remain to be implemented.
Ethics status	The revised eRA pre-screening questionnaire was routed to the supervisors on 20 August. It clarifies the use of public bibliographic metadata and synthetic corruptions, without personal-identifier linkage.

Questions guiding the review

- How are references extracted, resolved and matched to authoritative scholarly records?
- Which metadata fields provide reliable evidence when a citation is incomplete or corrupted?
- How should a tool communicate uncertainty, conflicting candidates and provenance?
- Where do existing services stop, and what workflow should this project add?

Draft literature review

1. Scholarly metadata as verification infrastructure

Reference checking depends on a reliable external record. Crossref exposes an open REST API over registered scholarly metadata and supports direct DOI lookup, bibliographic search and citation matching [1, 2]. DataCite offers a comparable public API for DOI metadata and provenance, while OpenAlex links works, authors, institutions, venues and topics in an open scholarly graph [3, 4]. These services make automated verification practical, but their records are not identical or complete. A checker therefore needs to preserve source provenance, compare fields explicitly and avoid treating the first search result as established fact.

Crossref identifies accurate bibliographic fields as central to citation display, DOI matching and discovery [5]. Comparative research also reports differences in metadata completeness among open scholarly databases [6]. The design implication is a source-aware record model. Each candidate should retain its provider, identifier, retrieved fields and retrieval time.

2. Extracting references from documents

Before a reference can be checked, its fields must be recovered from an unstructured string or document. ParsCit uses conditional random fields to identify reference strings, parse their components and locate citation contexts [7]. CERMINE extends the problem to scientific PDFs through a modular extraction pipeline and reports an average F-score of 77.5 percent across its extraction tasks [8]. These systems show that parsing errors are expected when layouts and citation styles vary. The verifier should keep the original string beside every parsed field so a user can inspect what the system inferred.

3. Record linkage and candidate ranking

A DOI provides the strongest route to a record because it can be resolved directly. References often arrive without one, so matching becomes an entity-resolution problem. Prior work combines title, author surnames, venue and year, with semantic similarity where literal matching is weak [9]. Recent pipeline work also treats incomplete metadata as a normal input [10]. A practical ranking model should therefore use several signals: normalized title similarity, author overlap, year distance, venue agreement and identifier agreement. Contradictory evidence should lower confidence even when titles appear similar.

The current prototype implements DOI lookup, title search and an initial title-similarity score. This is enough to test the interaction flow, but it is not yet a defensible confidence model. The next version will retain multiple candidates, expose individual feature scores and calibrate decision thresholds against labelled examples.

4. Validation, uncertainty and provenance

A useful verification tool must show why it reached a result. Work on OpenCitations validation distinguishes structural and semantic errors and emphasises traceable reporting [11]. The output should separate verified fields, probable mismatches, missing evidence and unresolved cases. A confidence label can summarize measured evidence, while an evidence panel exposes the record, source and comparison that produced it.

When candidates are close or metadata sources disagree, the tool should request human confirmation and retain the alternatives. Corrections should not silently overwrite the submitted reference. The original, proposed correction and supporting record should remain visible together.

5. Citation rendering and the project gap

Citation Style Language provides a mature, machine-readable way to render citations and bibliographies across thousands of styles [12]. Reusing CSL is more reliable than maintaining hand-written formatters. Existing metadata services, parsers, matching methods and style engines each solve part of the workflow. The project gap is the integration layer that takes a user's reference, retrieves plausible records, compares evidence at field level, reports uncertainty clearly and generates a traceable correction in the required style.

The main contribution will be an explainable validation workflow and its evaluation. Success should be measured on a frozen corpus of clean references and controlled corruptions, using candidate-retrieval accuracy, field-error detection, false-alarm rate, correction accuracy, response time and confidence calibration.

Design decisions taken from the review

Decision	Implementation consequence
Preserve evidence	Store the original reference, parsed fields, candidates, provider and retrieval timestamp.
Score several fields	Combine title, authors, year, venue and identifiers, and expose each signal.
Keep uncertainty visible	Use verified, warning and unresolved states. Ask for confirmation when evidence conflicts.
Evaluate reproducibly	Freeze a public-metadata corpus, create documented synthetic corruptions, and report accuracy by error type.
Use established rendering	Replace prototype formatters with a CSL processor after record verification.

Immediate work plan

- **Corpus:** Build a frozen set from public Crossref and DataCite records, then create labelled title, author, year, venue and DOI corruptions.
- **Service layer:** Normalize provider responses, retain all candidates and record provenance.
- **Matching:** Implement field-level features and a transparent weighted baseline before considering a learned ranker.
- **Evaluation:** Measure top-1 retrieval, field-error precision and recall, correction accuracy, latency and calibration.
- **Reporting:** Develop this draft into the final report chapter and record experimental decisions as they are made.

Source matrix

Literature cluster	What it establishes	Effect on this project
Crossref REST API and matching	Metadata retrieval, DOI lookup and bibliographic search	Core provider with DOI and title routes
OpenAlex	Open scholarly knowledge graph	Secondary candidates and coverage comparison
ParsCit and CERMINE	Reference parsing and document extraction	Preserve input and evaluate parser errors
Metadata matching studies	Multi-field candidate matching	Rank candidates using explainable features
OpenCitations validation	Structural and semantic checks	Field warnings and an evidence trace
Citation Style Language	Standardized reference rendering	Produce final correction output with CSL

References

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